Megaconial Congenital Muscular Dystrophy (CHKB): Comprehensive Disease Characterization Report

MONDO ID: MONDO:0011246 · **OMIM:** 602541 · **Causal gene:** *CHKB* (choline kinase beta, HGNC:1938, chromosome 22q13.33) · **Category:** Mendelian, autosomal recessive

Summary

Megaconial Congenital Muscular Dystrophy (MDCMC / MCMD) is an ultra-rare autosomal-recessive multisystem disorder caused by **biallelic loss-of-function variants in *CHKB*, the gene encoding choline kinase beta**. CHKB catalyzes the first, rate-limiting step of the Kennedy (CDP-choline) pathway for *de novo* biosynthesis of **phosphatidylcholine (PC)** — the most abundant phospholipid in eukaryotic membranes. Loss of CHKB activity disrupts PC-dependent mitochondrial membrane homeostasis and produces the disease's pathognomonic feature: **peripherally enlarged "megaconial" (giant) mitochondria with central mitochondrial depletion** on muscle biopsy. The causal gene was identified by Mitsuhashi et al. in 2011 (PMID: 21665002), building on the 1998 morphological description of a novel congenital muscular dystrophy with mitochondrial structural abnormalities (PMID: 9427222).

Clinically, the core phenotype comprises **congenital-to-early-childhood muscular dystrophy** (hypotonia, proximal weakness, muscle wasting, mildly elevated creatine kinase), **global developmental delay with severe intellectual disability and/or autistic features**, **ichthyosis-like skin changes**, and **dilated cardiomyopathy in approximately half of patients** — the leading cause of death. The phenotypic spectrum is broad and allelic, ranging from severe congenital dystrophy with profound intellectual disability to an adult-onset, non-progressive proximal myopathy with normal cognition ("Proximal Myopathy with Focal Depletion of Mitochondria"). Mechanistically, the PC-synthetic defect leads to mitochondrial membrane dysfunction, reduced mitochondrial fission (DRP1/MFF/FIS1), impaired fatty-acid β-oxidation, increased mitophagy, and bioenergetic failure (reduced Krebs-cycle intermediates and ATP).

No disease-modifying therapy is yet approved; management is **supportive** — physiotherapy, developmental/educational support, cardiac surveillance and treatment of dilated cardiomyopathy up to heart transplantation, and targeted dermatologic therapy (ustekinumab responded in one patient). The *Chkb*-deficient **rostrocaudal muscular dystrophy (rmd) mouse** faithfully recapitulates the disease and has enabled proof-of-concept **AAV6-*CHKB* gene-replacement therapy** and **CHKA (paralog) upregulation**, both of which rescue the dystrophic phenotype and now constitute the most promising preclinical therapeutic routes.

Section 1 — Disease Information

Overview. Megaconial Congenital Muscular Dystrophy is a form of congenital muscular dystrophy defined morphologically by giant ("megaconial") mitochondria at the periphery of muscle fibers with central sarcoplasmic mitochondrial depletion, and genetically by biallelic *CHKB* mutations. It was first delineated as a distinct clinical-pathological entity by Nishino et al. in 1998 in four patients from three unrelated families who had merosin-positive congenital muscular dystrophy with marked mental retardation (PMID: 9427222). The genetic cause (*CHKB*) was established in 2011 (PMID: 21665002).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0011246
OMIM	602541 (Muscular dystrophy, congenital, megaconial type / MDCMC)
Gene	*CHKB* (HGNC:1938; gene OMIM 612395; Ensembl ENSG00000100288)
Chromosomal locus	22q13.33 (head-to-head with *CPT1B*)
Orphanet	Megaconial congenital muscular dystrophy
ICD-10	G71.0 (muscular dystrophy, approximate)
MeSH	Muscular Dystrophies (no disease-specific term)

Synonyms / alternative names: Megaconial type congenital muscular dystrophy; Muscular dystrophy, congenital, megaconial type (MDCMC); CHKB-related congenital muscular dystrophy; Congenital muscular dystrophy, megaconial type; and the allelic **Proximal Myopathy with Focal Depletion of Mitochondria** (milder adult-onset form) (PMID: 26782016).

Data source type. The knowledge here is derived from **aggregated disease-level resources and published individual-patient case series/reports** (OMIM, Orphanet, and PubMed primary literature), not from electronic health records. Reported cohorts are small (largest series n=15; Iranian series n=13).

Section 2 — Etiology

Primary cause — genetic. The disease is caused exclusively by **biallelic (homozygous or compound heterozygous) loss-of-function variants in *CHKB***. Mitsuhashi et al. identified such mutations in 15 individuals with the characteristic clinical and morphological phenotype; in three individuals with nonsense mutations, choline kinase enzyme activity was undetectable and phosphatidylcholine levels were decreased (PMID: 21665002):

> "we have identified homozygous or compound heterozygous mutations in the gene encoding choline kinase beta (CHKB)... In muscle of three affected individuals with nonsense mutations, choline kinase activities were undetectable, and phosphatidylcholine levels were decreased."

Genetic risk factors. The causal variants are the sole genetic risk factor; there are no established susceptibility loci or proven human modifier genes. **Consanguinity** is a major contributing circumstance: most reported patients are homozygous (40 of 49 reported by 2023), consistent with autosomal-recessive inheritance and founder/consanguinity effects in populations with high rates of consanguineous marriage (Turkey, Iran, India, Japan, China) (PMID: 36896673; PMID: 37393748).

Environmental risk factors / protective factors / gene-environment interactions. None are established. As a monogenic Mendelian disorder, environmental exposures, lifestyle, toxins, and infectious agents are **not applicable** as causes. No protective genetic alleles or

environmental protective factors are documented. One molecularly relevant note: the paralogous gene **CHKA** can partially substitute for CHKB function (basis of a therapeutic strategy), which is a molecular redundancy rather than a gene-environment interaction.

Section 3 — Phenotypes

The best-characterized phenotype data come from the 15-patient series of Haliloglu et al. (PMID: 26067811); additional features come from other series and case reports.

> "Core clinical phenotype included global developmental delay prominent in gross-motor and language domains, severe intellectual disability (ID), and/or muscle weakness in all cases" (PMID: 26067811).

Phenotype	Type	Frequency (where reported)	Onset	Course	Suggested HPO
Global developmental delay	Clinical sign	15/15 (100%)	Neonatal–infancy	Static/slowly progressive	HP:0001263
Severe intellectual disability	Clinical sign	Majority (core feature)	Childhood	Static	HP:0010864
Muscle weakness (proximal)	Clinical sign	All cases (core)	Congenital–childhood	Slowly progressive/variable	HP:0003701
Muscle wasting / atrophy	Physical manifestation	Frequent	Childhood	Progressive	HP:0003202
Hypotonia	Clinical sign	Frequent	Neonatal	—	HP:0001252
Elevated serum creatine kinase (mild)	Laboratory abnormality	12/15 (80%)	—	—	HP:0003236
Ichthyosis-like / dry skin changes	Physical manifestation	11/15 (73%)	Childhood	Chronic	HP:0008064 / HP:0000958
Microcephaly	Physical manifestation	6/15 (40%)	Congenital	Static	HP:0000252
Dysmorphic features	Physical manifestation	7/15 (47%)	Congenital	Static	HP:0001999
Dilated cardiomyopathy	Clinical sign	~50% overall; 2/15 in series	Variable (childhood–adult)	Progressive	HP:0001644
Congenital heart defect	Clinical sign	3/15 (20%)	Congenital	—	HP:0001627
	Behavioral	Subset	Childhood	—	HP:0000729

Phenotype	Type	Frequency (where reported)	Onset	Course	Suggested HPO
Autistic behavior / ASD features					
Rhabdomyolysis susceptibility	Clinical sign	Mild (allelic) form	Adult	Episodic	HP:0003201

Severity and progression. Highly variable, including intrafamilial variability (PMID: 26067811). The dilated cardiomyopathy is the most clinically severe feature: "Clinical characteristics are delayed motor development, intellectual disability and dilated cardiomyopathy in half of reported cases" (PMID: 26855408).

Neurobehavioral overlap. CHKB disease can mimic **atypical Rett syndrome** (developmental regression, hand stereotypies) (PMID: 33712684) and present as **autism spectrum disorder** (PMID: 33623274) — important differential-diagnosis considerations.

Quality of life. No formal EQ-5D/SF-36/PROMIS studies exist. Qualitatively, the combination of severe intellectual disability, motor disability, and cardiac disease produces major impairment of daily functioning and lifelong dependency in the severe-end majority; the mild allelic form permits near-normal function.

Section 4 — Genetic / Molecular Information

Causal gene. *CHKB* (choline kinase beta), gene OMIM 612395; disease OMIM 602541. CHKB catalyzes **ATP + choline → ADP + phosphocholine** (EC 2.7.1.32), the first committed step of PC synthesis via the Kennedy pathway, and also phosphorylates ethanolamine. The protein functions as a homodimer/heterodimer with CHKA. Relevant CHEBI entities: choline, phosphocholine, phosphatidylcholine, ATP.

Pathogenic variants. All classes of loss-of-function variants are reported:

Variant type	Example(s)	Reference
Nonsense	Nonsense mutations with undetectable choline kinase activity	PMID: 21665002
Frameshift	c.598delC (p.Q200Rfs*11)	PMID: 39465137
Missense	c.1031G>A (p.R344Q); c.1060G>C (p.Gly354Arg); c.263C>T; c.950T>A	PMID: 27123443; PMID: 36175989; PMID: 26782016
Splice-site	c.1031+3G>C; c.818+1G>A; c.1031+1G>A; c.225-2A>T; c.448-56_29del (intronic)	PMID: 25740612; PMID: 33623274; PMID: 35707590; PMID: 36175989
Synonymous→splicing	c.843T>C (p.Phe281=) causing exon 8 skipping; c.581G>A causing exon 4 deletion	PMID: 41026188
Structural / UPD	Large heterozygous deletion with uniparental disomy masquerading as homozygosity	PMID: 36896673

Important molecular lesson: Several pathogenic *CHKB* variants act through **aberrant splicing**, including intronic and synonymous changes that escape standard coding-sequence analysis; RNA studies (RT-PCR from muscle or lymphocytes) are often needed to prove pathogenicity (PMID: 41026188; PMID: 25740612; PMID: 36175989). Truncated proteins are frequently degraded by nonsense-mediated decay.

Variant classification (ACMG/AMP): Reported variants are predominantly **pathogenic / likely pathogenic** with functional evidence (loss of enzyme activity, reduced DRP1, splicing assays). **Functional consequence:** uniform **loss of function** (no gain-of-function or dominant-negative mechanism).

Allele frequency. Causal alleles are ultra-rare / private in gnomAD, consistent with the disorder's rarity and recessive inheritance; many are population- or family-specific.

Somatic vs germline. Exclusively **germline**, biallelic.

Modifier genes / epigenetics / chromosomal abnormalities. No human modifier genes are confirmed. Of note, *CHKB* lies in a shared epigenetic regulatory domain head-to-head with *CPT1B* (carnitine palmitoyltransferase 1B), a key fatty-acid-oxidation gene; their coordinated regulation is relevant to the FAO defect (PMID: 33058877). No recurrent large chromosomal abnormalities beyond the single UPD/deletion case (PMID: 36896673).

Section 5 — Environmental Information

Not applicable as cause. Megaconial CMD is a monogenic disorder with no established environmental, lifestyle, toxic, or infectious contributors. No dietary, occupational, or exposure risk or protective factors are documented. Phosphatidylcholine/choline nutritional supplementation has been considered conceptually (given the biochemical defect) but is **not established** to modify disease, because the defect is in the enzymatic kinase step rather than in choline availability.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function *CHKB* mutation → leads to** loss of choline kinase beta enzyme activity in skeletal and cardiac muscle (demonstrated: choline kinase activity undetectable in nonsense-mutation muscle) (PMID: 21665002).
2. Loss of CHKB activity → **results in** a block at the first step of the Kennedy pathway (choline → phosphocholine) → **decreased de novo phosphatidylcholine (PC) synthesis** in muscle (demonstrated in patient muscle) (PMID: 21665002).
3. Reduced PC supply → **leads to** disturbed **mitochondrial membrane phospholipid homeostasis** (PC is a predominant mitochondrial membrane phospholipid) (PMID: 26855408; PMID: 22024749).
4. Membrane dysfunction → **results in** abnormal mitochondrial morphology: **peripheral giant ("megaconial") mitochondria with central sarcoplasmic depletion** — the pathognomonic lesion (PMID: 9427222).
5. **Branch 4a (fission defect):** reduced mitochondrial fission proteins **DRP1, MFF, FIS1** → impaired mitochondrial fission → enlarged mitochondria (demonstrated in patient skeletal muscle cells and tissue) (PMID: 34518586; PMID: 36175989).
6. **Branch 4b (mitophagy):** PC-synthetic defect → **increased mitophagy / mitochondrial dysfunction** (PMID: 22024749).
7. Mitochondrial dysfunction → **leads to impaired fatty-acid β-oxidation**, with shunting of fatty acids into triacylglycerol and decreased PPAR/target-gene expression (demonstrated in

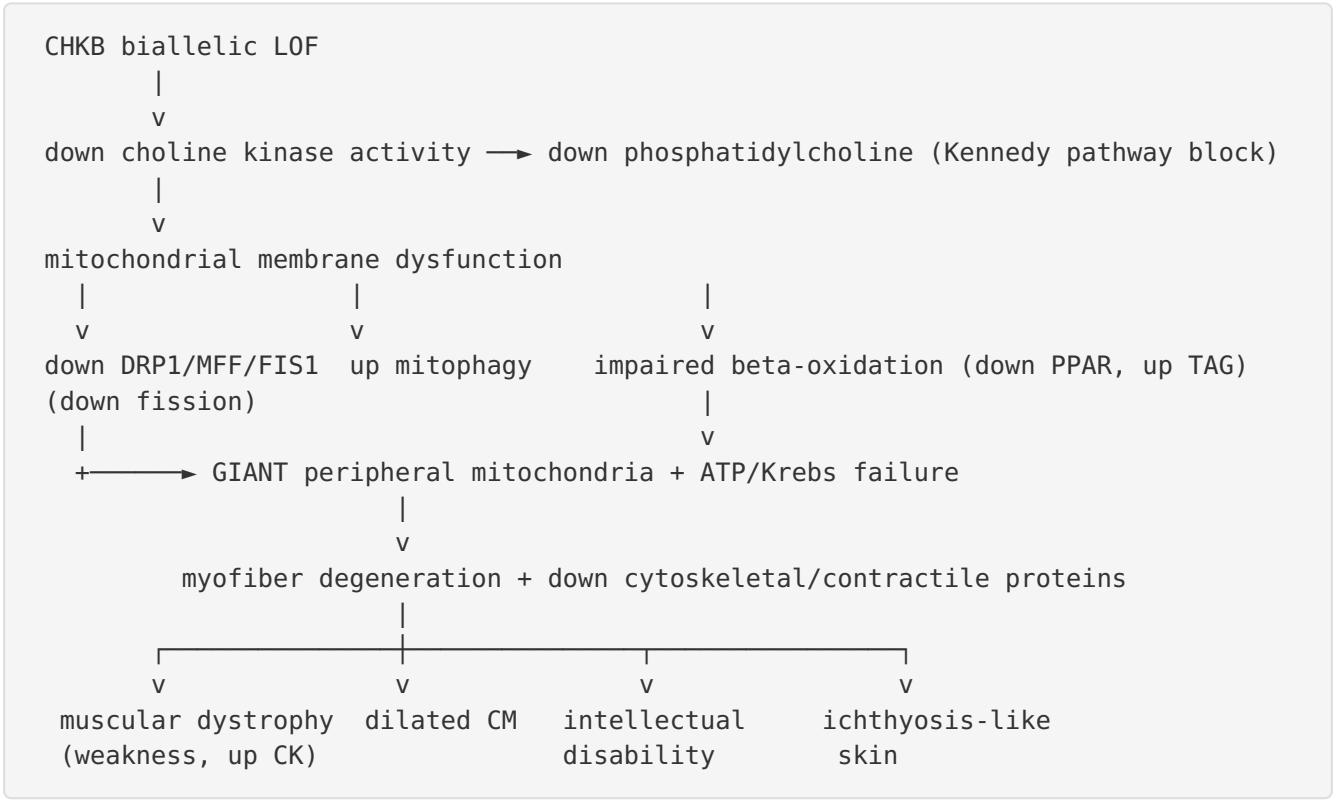
Chkb mouse; notably PC level is not significantly altered in that model, implicating a temporal metabolic mechanism) (PMID: 35322809).

8. Impaired β-oxidation + reduced OXPHOS (mild complex decreases, incomplete complex V assembly) → **results in bioenergetic failure**: decreased phosphonucleotides, Krebs-cycle intermediates, and ATP (PMID: 34518586; PMID: 26855408).
9. Energy failure + membrane instability → **leads to** myofiber degeneration/necrosis and regeneration (dystrophic pathology), plus **downregulation of cytoskeletal and contractile proteins** (proteomics: 218 differentially abundant proteins, mostly reduced) (PMID: 42449274; PMID: 9427222).
10. → **results in** the clinical manifestations: **skeletal muscle weakness/dystrophy**, **dilated cardiomyopathy** (cardiac hypertrophy and arrhythmia predisposition in mouse), **CNS dysfunction** (intellectual disability, developmental delay — mechanism inferred, less well demonstrated), and **ichthyosis-like skin** (inferred from PC's role in epidermal barrier lipids).

Supporting mechanistic detail

- **Molecular pathways:** Kennedy / CDP-choline pathway for PC biosynthesis (KEGG hsa00564 glycerophospholipid metabolism); mitochondrial fatty-acid β-oxidation; PPAR signaling.
- **Cellular processes (GO):** mitochondrial fission (GO:0000266), mitophagy (GO:0000423), fatty-acid beta-oxidation (GO:0006635), phosphatidylcholine biosynthetic process (GO:0006656), myofiber apoptosis/necrosis.
- **Protein dysfunction:** loss of CHKB catalytic function (choline kinase activity GO:0004103); secondary reduction of fission GTPase DRP1.
- **Metabolic changes:** reduced PC (patient muscle); impaired FAO with triacylglycerol accumulation; reduced ATP and Krebs-cycle intermediates — an energy-metabolism signature.
- **Tissue damage mechanisms:** bioenergetic insufficiency and membrane instability drive myonecrosis and fibrosis/connective-tissue increase.
- **Immune involvement:** none intrinsic; the skin phenotype may involve the IL-12/IL-23 axis (basis for ustekinumab efficacy) (PMID: 39143029).

ASCII causal diagram



Molecular profiling evidence: Proteomics of patient myoblasts (cytoskeletal/contractile downregulation; PMID: 42449274); metabolomics/fluxomics of patient muscle cells (bioenergetic signature; PMID: 34518586); transcriptomic/lipid data from the *Chkb* mouse (PMID: 35322809).

Section 7 — Anatomical Structures Affected

Organ level. - **Primary:** skeletal muscle (UBERON:0001134) — the principal affected organ. - **Secondary:** heart / myocardium (UBERON:0000948, UBERON:0002349) — dilated cardiomyopathy in ~50%; brain (UBERON:0000955) — intellectual disability/developmental delay; skin / epidermis (UBERON:0002097, UBERON:0001003) — ichthyosis-like changes. - **Body systems:** musculoskeletal, cardiovascular, nervous (CNS), integumentary.

Tissue and cell level. - Muscle tissue (UBERON:0002385); **skeletal muscle fiber / myocyte** (CL:0000188 skeletal muscle cell; CL:0008002 skeletal muscle fiber); cardiomyocyte (CL:0000746); epidermal keratinocyte (CL:0000312). Organelle disruption was shown to be **specific to skeletal muscle cells grown in culture** (PMID: 34518586).

Subcellular level. - **Mitochondrion (GO:0005739)** — the central affected compartment: peripherally enlarged mitochondria with complicated cristae and central depletion (PMID: 9427222). Mitochondrial inner/outer membranes (GO:0005743 / GO:0005741). Secondary involvement of the endoplasmic reticulum (site of Kennedy-pathway PC synthesis, GO:0005789) is inferred.

Localization / lateralization. Generalized, **bilateral and symmetric** muscle involvement with proximal predominance; no lateralization.

Section 8 — Temporal Development

Onset. Typically **congenital to early childhood** — neonatal hypotonia, infantile developmental delay; "a newly defined rare autosomal recessive disorder, with multisystem involvement presenting from the neonatal period to adolescence" (PMID: 37393748). Onset pattern is **insidious/chronic**. The milder allelic form (Proximal Myopathy with Focal Depletion of Mitochondria) has **adult onset** (PMID: 26782016).

Progression. Slowly progressive muscle weakness/wasting; intellectual disability is largely static. Dilated cardiomyopathy may develop or progress over childhood–adolescence and can cause heart failure and death. Disease duration is **chronic/lifelong**.

Course pattern. Predominantly **slowly progressive**; the mild allelic form is **non-progressive** but with **episodic rhabdomyolysis** susceptibility (PMID: 26782016).

Patterns / critical periods. No spontaneous remission. The chief window for intervention is before onset of significant cardiac disease (cardiac surveillance) and, for emerging gene therapy, potentially after disease onset — AAV6-*Chkb* rescued the mouse even when given **post-disease onset** (PMID: 31216357).

Section 9 — Inheritance and Population

Inheritance. Autosomal recessive (biallelic *CHKB* loss of function). Penetrance appears **complete** for biallelic LOF; **expressivity is variable**, including intrafamilial variability (PMID: 26067811). No genetic anticipation. One documented case of apparent homozygosity resulted from a large heterozygous deletion plus **uniparental disomy** (PMID: 36896673).

Epidemiology. Ultra-rare; precise prevalence/incidence unknown. **~35 confirmed patients** recorded by 2021 (PMID: 33712684); ***CHKB* mutations described in 49 patients by 2023, of whom 40 were homozygous** (PMID: 36896673):

> "CHKB mutations have been described in 49 patients with megaconial congenital muscular dystrophy, which is a rare autosomal recessive disorder, of which 40 patients showed homozygosity" (PMID: 36896673).

Population / geography. Cases are reported across **Turkey, Japan, Iran (13-patient series), India, China, and Europe (Italy, Spain, Portugal)**, frequently in **consanguineous families** (PMID: 37393748; PMID: 26067811). High homozygosity indicates **founder/consanguinity effects**. No strong sex bias is established (autosomal). Carrier frequency is not formally estimated but is presumed very low outside consanguineous populations.

> "a newly defined rare autosomal recessive disorder, with multisystem involvement presenting from the neonatal period to adolescence" (PMID: 37393748).

Section 10 — Diagnostics

Diagnostic hallmark — muscle biopsy. Myopathic/dystrophic changes with the **pathognomonic peripheral enlarged ("megaconial") mitochondria and central sarcoplasmic mitochondrial depletion**, best shown on **modified Gomori trichrome stain**, oxidative enzyme histochemistry, and **electron microscopy** (PMID: 33712684; PMID: 9427222; PMID: 37393748):

> "Muscle biopsy in two patients showed a myopathic pattern with characteristic peripherally placed enlarged mitochondria on modified Gomori trichrome stain and electron microscopy" (PMID: 33712684).

Laboratory. Serum creatine kinase is **mildly elevated** (e.g., up to ~500 U/L; 12/15 patients) — a useful but non-specific clue (PMID: 26067811; PMID: 27123443). LOINC: CK [Creatine kinase, serum].

Imaging / electrophysiology. Brain MRI is typically **normal** (PMID: 36175989); EMG may be normal or myopathic. Echocardiography/ECG are essential for cardiac surveillance (dilated cardiomyopathy, conduction defects — e.g., fascicular/bundle-branch block) (PMID: 39465137).

Genetic testing (confirmatory). Whole-exome sequencing or targeted **CHKB single-gene testing / congenital-muscular-dystrophy gene panels** confirm the diagnosis (PMID: 25740612; PMID: 33712684). Because several pathogenic variants are **intronic/synonymous splice-altering**, **RNA studies (RT-PCR) and careful reanalysis** are sometimes required (PMID: 41026188; PMID: 25740612). Chromosomal microarray can detect large deletions/UPD (PMID: 36896673). Mitochondrial DNA is normal (normal mtDNA content by in situ hybridization) (PMID: 9427222).

Alternative cell source. Giant mitochondria can be detected in **lymphocytes/other cells when muscle is unavailable** (PMID: 36896673):

> "We provide a possibility to detect giant mitochondria in other cells when muscle was not available."

Clinical clue — skin. Ichthyosis-like/psoriasiform skin changes with pruritus and nummular eczema may point to the diagnosis among congenital muscular dystrophies (PMID: 27123443; PMID: 37907281).

Differential diagnosis. Other congenital/merosin-positive muscular dystrophies, mitochondrial myopathies, atypical Rett syndrome, autism spectrum disorder, and hypotonia-cystinuria syndrome (a case co-occurred/mimicked) (PMID: 33712684; PMID: 33623274; PMID: 35707590).

Screening. No newborn screening exists. **Carrier and cascade testing** within affected consanguineous families, and prenatal/preimplantation testing, are feasible once the familial variants are known.

Section 11 — Outcome / Prognosis

Mortality. The principal cause of death is **dilated cardiomyopathy / heart failure**; a patient in the original 1998 series died of dilated cardiomyopathy at age 13 (PMID: 9427222). Cardiac disease can require **heart transplantation** (PMID: 26855408). Formal survival/life-expectancy statistics are not available given rarity.

Morbidity / disability. In the severe-end majority: lifelong **severe intellectual disability**, motor disability, and dependency; variable cardiac and dermatologic morbidity. The mild allelic form permits near-normal lifespan and function with episodic rhabdomyolysis risk (PMID: 26782016).

Complications. Progressive weakness/contractures, cardiac failure/arrhythmia, respiratory compromise (inferred in severe cases), feeding/growth difficulties, and skin complications.

Prognostic factors. The **molecular defect (genotype) influences clinical outcome** — e.g., nonsense/complete-LOF alleles associate with severe disease, whereas some missense/hypomorphic or partial-splicing alleles yield milder phenotypes; the clinical outcome "seems to be influenced by CHKB molecular defects" (PMID: 36175989). The presence and severity of dilated cardiomyopathy is the dominant prognostic determinant.

Section 12 — Treatment

No approved disease-modifying therapy exists; management is supportive and multidisciplinary.

Supportive / rehabilitative (NCIT: Supportive Care; Physical Therapy; Occupational Therapy): - Physiotherapy, occupational and speech therapy; management of contractures; developmental and educational support (PMID: 26855408). - **Cardiac surveillance and heart-failure management**, up to **heart transplantation** for severe dilated cardiomyopathy (PMID: 26855408): "who was treated by heart transplantation." (NCIT: Heart Transplantation.)

Dermatologic (targeted). The IL-12/IL-23 inhibitor **ustekinumab** produced a successful response in a patient with diffuse ichthyosis-like scaling (PMID: 39143029):

> "We report a patient with MCMD who had diffuse ichthyosis-like scaling, and successfully responded to ustekinumab" (PMID: 39143029). (NCIT: Ustekinumab.)

Advanced / experimental (preclinical). - **AAV-mediated CHKB gene replacement:** Intramuscular **AAV6-*Chkb*** given **after disease onset** rescued the dystrophic phenotype in the *rmd* mouse; a muscle-specific *Chkb* transgene completely rescued motor/behavioral function (PMID: 31216357):

> "Intramuscular gene therapy post-disease onset using an adeno-associated viral 6 (AAV6) vector carrying a functional copy of Chkb is also capable of rescuing the dystrophy phenotype" (PMID: 31216357). - **CHKA (paralog) upregulation:** AAV6-driven upregulation of the paralog **Chka** also improved phenotypes — an alternative genetic-rescue strategy (PMID: 31216357). - Continued **preclinical gene-therapy development** (PMID: 42292914) and mechanism-based therapeutic routes targeting the lipid-metabolism defect (PMID: 35322809).

Pharmacogenomics / combination / personalized medicine. None established. **Genotype-guided prognosis** is informative (see Section 11). No approved small-molecule or RNA therapy.

Section 13 — Prevention

Primary prevention is genetic, given the Mendelian recessive etiology: - **Genetic counseling** for consanguineous families and families with an affected child (recurrence risk 25%). - **Carrier/cascade screening** once the familial variant is identified; **prenatal diagnosis** and **preimplantation genetic testing** are feasible. - **Secondary prevention / early detection:** early echocardiographic cardiac surveillance to detect and manage dilated cardiomyopathy before heart failure; early developmental intervention. - **Tertiary prevention:** physiotherapy to prevent contractures; heart-failure therapy; dermatologic management.

No vaccination, environmental, or public-health interventions are applicable. There is no population newborn screening for this disorder.

Section 14 — Other Species / Natural Disease

- **Taxonomy / orthologs:** Human *CHKB* (NCBI Gene 1120) has a mouse ortholog *Chkb* (*Mus musculus*, NCBI Taxon 10090).
- **Natural animal disease:** The **rostrocaudal muscular dystrophy (rmd) mouse** carries a naturally occurring *Chkb* deletion and develops an MDCMC-like dystrophy — the key natural/model disease (PMID: 31216357). No well-characterized companion-animal natural disease is documented in the reviewed literature.
- **Comparative biology / conservation:** The CHKB–PC–mitochondrial mechanism is conserved; *Chkb*-null mice reproduce muscle and cardiac disease, supporting evolutionary conservation. The *CHKB–CPT1B* syntenic/regulatory linkage is a eutherian-mammal feature relevant to muscle fatty-acid oxidation and brown-adipose-tissue evolution (PMID: 33058877).
- **Transmission / zoonosis:** Not applicable (genetic disease).

Section 15 — Model Organisms

Primary model — mouse. The ***Chkb*-deficient rostrocaudal muscular dystrophy (rmd) mouse** is the central model.

Model aspect	Detail	Reference
Model type	Mammalian (mouse), naturally occurring *Chkb* deletion	PMID: 31216357
Phenotype recapitulation	MDCMC-like muscular dystrophy; motor/behavioral deficits; cardiac hypertrophy and arrhythmia predisposition	PMID: 31216357; PMID: 35151687
Metabolic phenotype	Temporal failure of mitochondrial β-oxidation; shunting into triacylglycerol; reduced PPAR targets (PC level not significantly altered — a model nuance)	PMID: 35322809
Therapeutic validation	Muscle-specific *Chkb* transgene = complete rescue; AAV6-*Chkb* post-onset = rescue; AAV6-*Chka* upregulation = improvement	PMID: 31216357

Cellular / in vitro models. Patient-derived primary skeletal myoblasts/muscle cells recapitulate reduced mitochondrial fission (↓DRP1/MFF/FIS1), impaired energy metabolism, and cytoskeletal/contractile proteomic alterations (PMID: 34518586; PMID: 42449274); notably, organelle disruption was **specific to skeletal muscle cells in culture**.

Model limitations. The mouse does not fully model the human CNS/intellectual-disability phenotype; and the discrepancy that **PC levels are not significantly reduced in the *Chkb* mouse** (whereas human nonsense-mutation muscle shows decreased PC) indicates the mouse may capture downstream metabolic consequences better than the primary PC-deficiency readout — relevant when interpreting mechanism and therapeutic endpoints.

Resources: MGI (*Chkb*), IMPC/IMSR for mouse lines; patient-derived myoblast cultures.

Mechanistic Model / Interpretation

Megaconial CMD is best understood as a **phospholipid-biosynthesis disorder with a downstream mitochondrial/bioenergetic phenotype**. The single necessary and sufficient lesion is **biallelic CHKB loss of function**, which removes the first enzymatic step of the Kennedy pathway and depletes phosphatidylcholine supply to muscle mitochondrial membranes. The disease's signature — **giant peripheral mitochondria with central depletion** — arises from two converging consequences: (i) **impaired mitochondrial fission** due to reduced DRP1/MFF/FIS1, and (ii) **increased mitophagy** with abnormal membrane dynamics. The functional cost is an **energy crisis**: impaired fatty-acid β-oxidation (with lipid shunting into triacylglycerol), mild OXPHOS/complex-V deficits, and reduced ATP and Krebs-cycle intermediates. The muscle then undergoes dystrophic degeneration with secondary loss of cytoskeletal/contractile proteins, producing weakness and elevated CK. The same mechanism injures cardiomyocytes (dilated cardiomyopathy — the principal lethal complication) and, via less-well-demonstrated routes, the brain (intellectual disability) and skin (ichthyosis). Genotype grades the outcome: complete-LOF alleles → severe congenital disease; hypomorphic/partial-splicing alleles → milder, even adult-onset, phenotypes.

The therapeutic logic follows directly: restoring CHKB function (AAV gene replacement) or boosting the redundant paralog CHKA re-establishes PC synthesis and rescues the mouse, making **gene-directed therapy the leading translational path**, while current care remains supportive with genotype-informed prognosis and cardiac surveillance as the highest-yield clinical interventions.

Evidence Base

PMID	Title (abbrev.)	Contribution
21665002	*CMD caused by defective de novo PC biosynthesis*	**Landmark:** identifies *CHKB* cause; absent choline kinase activity, reduced PC
9427222	*New CMD with mitochondrial structural abnormalities*	First morphological description of megaconial mitochondria; normal mtDNA
26067811	*Clinical characteristics, 15 patients*	Core phenotype frequencies; intrafamilial variability
26855408	*Mitochondrial membrane homeostasis defect*	DCM in half of cases; OXPHOS/complex-V findings; heart transplant
34518586	*Reduced fission & impaired energy metabolism*	↓ DRP1/MFF/FIS1; bioenergetic signature in patient cells
35322809	*Mechanism & therapeutic route (lipid metabolism)*	FAO failure, TAG shunting; PC not altered in mouse
22024749	*Phospholipid synthetic defect & mitophagy*	Links PC defect to mitochondrial dysfunction + mitophagy
31216357	*Functional rescue in mouse model*	AAV6-*Chkb* and *Chka* rescue; transgene rescue
35151687	*Mouse cardiac abnormalities / arrhythmia*	Cardiac hypertrophy and arrhythmia predisposition
26782016	*Proximal myopathy with focal mito depletion*	Mild allelic adult-onset end of spectrum
33712684	*Resemble atypical Rett syndrome*	Biopsy hallmark; Rett-like presentation; ~35 patients
33623274	*Rare cause of ASD*	Autism spectrum presentation; splice variants
36896673	*Large deletion + UPD*	49 patients/40 homozygous; lymphocyte detection
37393748	*13 Iranian patients*	AR, multisystem, neonatal–adolescent onset
42449274	*Proteomic analysis of myoblasts*	Cytoskeletal/contractile downregulation (218 proteins)
39143029	*Ustekinumab response*	Targeted therapy for cutaneous phenotype

PMID	Title (abbrev.)	Contribution
27123443	*Skin changes in differential diagnosis*	p.R344Q; ichthyosis as diagnostic clue
36175989	*Novel CHKB variants, review*	Intronic splice variant; ↓DRP1; genotype–outcome link
25740612	*Splicing mutation by WES*	c.1031+3G>C splice variant; WES utility
41026188	*Alternative splicing of 3 rare variants*	Synonymous/intronic variants alter splicing
39465137	*Rare homozygous variant*	c.598delC frameshift; severe cardiac/conduction disease
33058877	*CHKB-CPT1B coordination*	Shared regulatory domain; FAO/muscle relevance
42292914	*Preclinical gene therapy*	Ongoing AAV-CHKB gene-therapy development

Limitations and Knowledge Gaps

1. **Small evidence base:** Fewer than ~50 genetically confirmed patients worldwide; most data are case reports/small series, limiting frequency and natural-history estimates.
2. **No formal epidemiology:** Prevalence/incidence, carrier frequency, sex ratio, and survival statistics are not rigorously established.
3. **CNS mechanism unclear:** The pathway from CHKB/PC deficiency to intellectual disability and autistic/Rett-like features is inferred, not mechanistically demonstrated.
4. **Human vs mouse discrepancy:** PC is reduced in human nonsense-mutation muscle but *not* significantly altered in the *Chkb* mouse, complicating biomarker selection and mechanistic interpretation.
5. **No approved therapy and no human trials yet:** Gene-therapy evidence is entirely preclinical; ustekinumab benefit rests on a single case.
6. **Genotype–phenotype correlations** are suggested but not systematically quantified across variant classes.

7. **Variant detection pitfalls:** Deep-intronic, synonymous, and structural (deletion/UPD) variants can be missed without RNA studies and CMA.
8. **No disease-specific QoL instruments** validated for this disorder.

Proposed Follow-up Experiments / Actions

1. **International patient registry & natural-history study** to quantify prevalence, cardiac penetrance/age-of-onset, survival, and genotype–phenotype correlations.
2. **IND-enabling AAV-*CHKB* gene-therapy program:** optimize vector/serotype, dose, route (systemic vs intramuscular), and timing; define regulatory endpoints given the mouse PC-biomarker nuance (PMID: 31216357; PMID: 42292914).
3. **Test CHKA-upregulation / small-molecule approaches** as a non-vector-limited alternative (PMID: 31216357).
4. **CNS-focused modeling** (patient iPSC-derived neurons/organoids; brain-targeted mouse studies) to define the intellectual-disability mechanism.
5. **Biomarker development:** validate muscle PC, lipidomic FAO/TAG signatures, DRP1/fission markers, and circulating/imaging markers for diagnosis and trial endpoints (PMID: 34518586; PMID: 42449274).
6. **Prospective cardiac surveillance protocol** (serial echo/ECG/MRI) to guide early heart-failure intervention and transplant timing.
7. **Systematic splicing/RNA diagnostic workflow** and CMA to capture intronic/synonymous/structural variants that escape standard panels (PMID: 41026188; PMID: 36896673).
8. **Controlled evaluation of ustekinumab / IL-23-axis therapy** for the cutaneous phenotype beyond the single reported case (PMID: 39143029).

Report compiled from 27 reviewed publications and 9 confirmed findings. Evidence types span human clinical case series/reports, mouse-model studies, patient-derived in vitro work, and molecular/biochemical assays, as indicated per claim.
